

Epic 1: Define Problem and Understanding

Story 1: Identify and define the agricultural problem that the crop recommendation system aims to solve.

Story 2: Gather and analyze business requirements to understand project objectives and expected outcomes.

Story 3: Conduct a literature survey to study existing crop recommendation techniques and machine learning approaches.

Story 4: Analyze the social and business impact of implementing an AI-driven crop recommendation solution.

Epic 2: Data Collection and Analysis

Story 1: Download and collect the agricultural dataset from reliable sources.

Story 2: Import the required Python libraries for data analysis, visualization, and machine learning.

Story 3: Read and explore the dataset to understand its structure, features, and target variables.

Story 4: Perform univariate analysis to examine the distribution and characteristics of individual features.

Story 5: Conduct bivariate analysis to identify relationships between agricultural parameters and crop suitability.

Story 6: Perform multivariate analysis to discover patterns and interactions among multiple variables.

Epic 3: Data Pre-Processing

Story 1: Check for null values and handle missing data to improve dataset quality.

Story 2: Detect and treat outliers that may affect model performance.

Story 3: Extract seasonal crop information and prepare relevant features for analysis.

Story 4: Split the dataset into training and testing sets for model development and evaluation.

Epic 4: Model Building

Story 1: Apply K-Means Clustering to identify patterns and group similar agricultural conditions.

Story 2: Train a Logistic Regression model to predict suitable crops based on environmental parameters.

Story 3: Evaluate model performance using appropriate metrics and compare results.

Story 4: Save the best-performing model for deployment and generate crop predictions based on user-provided inputs.

Epic 5: Application Building

Story 1: Design and develop HTML pages to create an interactive and user-friendly interface.

Story 2: Build the Python backend code and integrate it with the trained machine learning model.

Story 3: Run, test, and validate the application to ensure accurate crop recommendations and smooth functionality.